```
from google.colab import files
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imread(filename)
print("Image uploaded successfully:", filename)

# Convert to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply dilation
dilated_image = cv2.dilate(gray, kernel, iterations=1)

# Display original image
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")
plt.show()

# Display original and dilated images side by side
plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(dilated_image, cmap="gray")
plt.title("Dilated Image")
plt.axis("off")

plt.show()
```

Choose Files Screenshot… 075823.png

Screenshot 2026-10-10 075823.png(image/png) - 93416 bytes, last modified: 10/10/2026 - 100% done

```
Saving Screenshot 2026-10-10 075823.png to Screenshot 2026-10-10 075823.png
Image uploaded successfully: Screenshot 2026-10-10 075823.png
```

Original Image



Original Image



Dilated Image



What is Colab?

Colab, or "Colaboratory", allows you to write and execute Python in your browser, with

- Zero configuration required
- Access to GPUs free of charge
- Easy sharing

Whether you're a **student**, a **data scientist** or an **AI researcher**, Colab can make your work easier. Watch Introduction to Colab or Colab Features You May Have Missed to learn more, or just get started below!

Getting started

The document you are reading is not a static web page, but an interactive environment called a **Colab notebook** that lets you write and execute code.

For example, here is a **code cell** with a short Python script that computes a value, stores it in a variable, and prints the result:

```
seconds_in_a_day = 24 * 60 * 60
seconds_in_a_day
```

```
86400
```